

MAT 110 Differential Calculus & Coordinate Geometry

→ Lagrange Multiplier

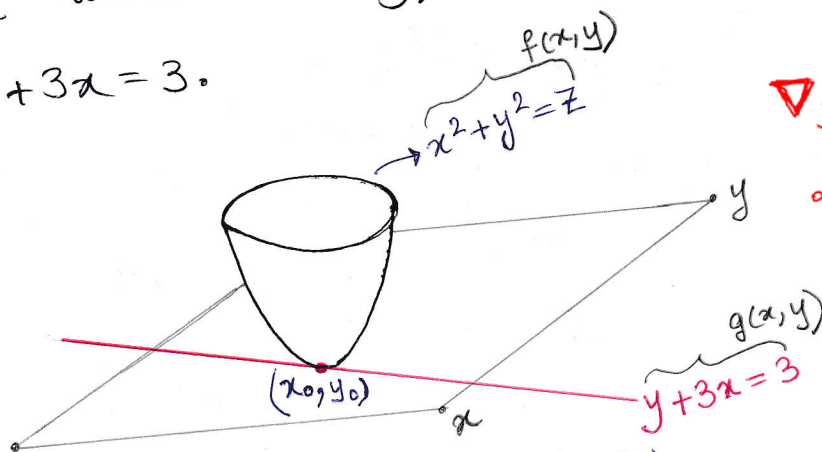
Lagrange Multiplier:

Find maxima/minima of $f(x,y)$ subject to
constraint $g(x,y) = c$
restriction/condition

For example we have $f(x,y) = x^2 + y^2$ and constraint

$$g(x,y) = y + 3x = 3.$$

$$\begin{aligned} \square: x \\ \square\square: y \end{aligned} \left. \vphantom{\begin{aligned} \square: x \\ \square\square: y \end{aligned}} \right\} x \propto y$$
$$y = 2x$$
$$x = \frac{1}{2}y$$



$$\nabla f \propto \nabla g$$

$$\therefore \nabla f = \lambda \nabla g$$

λ = scalar number

$(x_0, y_0) \rightarrow$ critical point

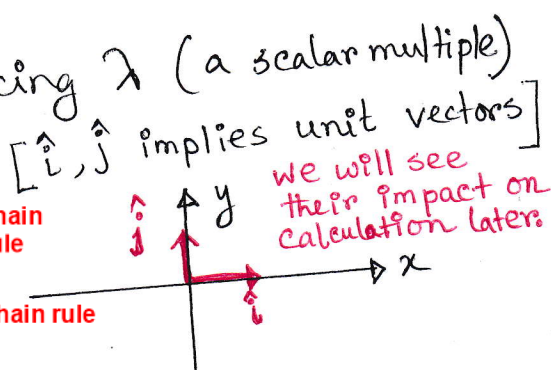
We will decide whether extremum occurs at (x_0, y_0) or not.
maximum/minimum specifically

$$\nabla g \neq 0 \quad \nabla \rightarrow \text{slope/gradient on xy plane}$$

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0) \quad \text{introducing } \lambda \text{ (a scalar multiple)}$$

$$\text{while } \nabla f(x,y) = \frac{df}{dx} \hat{i} + \frac{df}{dy} \hat{j} \rightarrow \text{chain rule}$$

$$\nabla g(x,y) = \frac{dg}{dx} \hat{i} + \frac{dg}{dy} \hat{j} \rightarrow \text{chain rule}$$



Chapter 13.9 pg 996 Anton's Calculus 10th Ed.

Exercise:

Use Lagrange multipliers to find the maximum & minimum values of f subject to the given constraint. Also, find the pts at which these extreme values occur

⑤ $f(x, y) = xy$; $\underbrace{4x^2 + 8y^2 = 16}_{g(x, y)}$

$$\nabla f = y\hat{i} + x\hat{j} \quad \nabla g = 8x\hat{i} + 16y\hat{j}$$

At a constraint, ^{for} relative extremum we must have:

$$\nabla f = \lambda \nabla g$$

$$\Rightarrow y\hat{i} + x\hat{j} = \lambda (8x\hat{i} + 16y\hat{j})$$

Equating factors of Like terms such as
(coefficients of $\hat{i}$ & coefficients of $\hat{j}$)

$$y = \lambda 8x \quad \text{--- (i)}$$

$$x = \lambda 16y \quad \text{--- (j)}$$

$$\lambda = \frac{y}{8x}$$

$$\lambda = \frac{x}{16y}$$

compare

$$\frac{y}{8x} = \frac{x}{16y}$$

$$\frac{y}{x} = \frac{x}{2y}$$

$$\boxed{x^2 = 2y^2} \quad \text{--- (2)}$$

$$\boxed{y^2 = \frac{1}{2} x^2} \quad \text{--- (1)}$$

Substitute y^2 into $g(x, y)$ } Alternatively sub x^2 into $g(x, y)$.

$$4x^2 + 8\left(\frac{1}{2}x^2\right) = 16$$

$$4x^2 + 4x^2 = 16$$

$$8x^2 = 16$$

$$x^2 = 2$$

$$x = \pm\sqrt{2}$$

$$4(2y^2) + 8y^2 = 16$$

$$16y^2 = 16$$

$$y^2 = 1$$

$$y = \pm 1$$

Sub $y = \pm 1$ into (2)

$$x^2 = 2(\pm 1)^2 = 2$$

$$x = \pm\sqrt{2}$$

Substitute $x = \pm\sqrt{2}$ into (1)

$$y^2 = \frac{1}{2}(\pm\sqrt{2})^2 = \frac{1}{2}(2) = 1$$

$$\therefore y = \pm\sqrt{1} = \pm 1$$

All possible combination of (x, y) :

(x, y)	$(-\sqrt{2}, -1)$	$(\sqrt{2}, 1)$	$(-\sqrt{2}, 1)$	$(\sqrt{2}, -1)$
$f(x, y)$ $= xy$	$(\sqrt{2})(-1)$ $= \sqrt{2}$	$(\sqrt{2})(1)$ $= \sqrt{2}$	$(-\sqrt{2})(1)$ $= -\sqrt{2}$	$(\sqrt{2})(-1)$ $= -\sqrt{2}$
max/min	max	max	min	min

$\rightarrow f(x, y)$ has maxima at $(-\sqrt{2}, -1)$ and $(\sqrt{2}, 1)$

The maximum $= \sqrt{2}$

$\rightarrow f(x, y)$ has minima at $(-\sqrt{2}, 1)$ and $(\sqrt{2}, -1)$

The minimum $= -\sqrt{2}$

$$(6) f(x, y) = x^2 - y^2; \quad g(x, y) = x^2 + y^2 - 25 = 0$$

$$\nabla f = 2x\hat{i} - 2y\hat{j} \quad \nabla g = 2x\hat{i} + 2y\hat{j}$$

$$\nabla f = \lambda \nabla g$$

$$2x\hat{i} - 2y\hat{j} = \lambda (2x\hat{i} + 2y\hat{j})$$

Equating factors of like terms:

$$2x = \lambda 2x \quad \text{--- (i)} \quad -2y = \lambda 2y \quad \text{--- (j)}$$

$$2x - \lambda 2x = 0$$

$$0 = 2y\lambda + 2y$$

$$2x(1 - \lambda) = 0$$

$$2y(\lambda + 1) = 0$$

$$2x = 0$$

$$2y = 0$$

$$x = 0$$

$$y = 0$$

Substitute either $x = 0$ or $y = 0$ into $g(x, y)$

Substituting $y = 0$ into $g(x, y)$

$$x = 0 \text{ @ } g(x, y)$$

$$y^2 = 25$$

$$y = \pm 5$$

$$(0, \pm 5)$$

$$x^2 + 0^2 - 25 = 0$$

$$x^2 = 25$$

$$x = \pm 5$$

All possible combination of (x, y) :

(x, y)	$(5, 0)$	$(-5, 0)$	$(0, 5)$	$(0, -5)$
$f(x, y)$ $= x^2 - y^2$	$5^2 - 0^2$ $= 25$	$(-5)^2 - 0^2$ $= 25$	$0^2 - 5^2$ $= -25$	$0^2 - (-5)^2$ $= -25$
max/min	max	max	min	min

⑦ $f(x,y) = 4x^3 + y^2$; $\underbrace{2x^2 + y^2 - 1}_{g(x,y)} = 0$

$$\nabla f = 12x^2 \hat{i} + 2y \hat{j}$$

$$\nabla g = 4x \hat{i} + 2y \hat{j}$$

$$12x^2 \hat{i} + 2y \hat{j} = \lambda (4x \hat{i} + 2y \hat{j})$$

Equating factors of like terms:

$$12x^2 = 4x\lambda \quad \text{--- (i)}$$

$$3x^2 = x\lambda$$

$$3x^2 - x\lambda = 0$$

$$x(3x - \lambda) = 0$$

$$x = 0, \quad 3x - \lambda = 0$$

$$x = \frac{\lambda}{3} = \frac{1}{3}$$

$$\therefore x = 0, \frac{1}{3}$$

$$2y = \lambda 2y \quad \text{--- (j)}$$

$$2y - \lambda 2y = 0$$

$$2y(1 - \lambda) = 0$$

$$2y = 0, \quad 1 - \lambda = 0$$

$$\lambda = 1$$

$$y = 0$$

substitute $y = 0$ into $g(x,y)$

$$2x^2 + 0^2 - 1 = 0$$

$$2x^2 = 1$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

$x = 0$ into $g(x,y)$

$$2(0)^2 + y^2 - 1 = 0$$

$$y = \pm 1 \quad (0, \pm 1)$$

$x = \frac{1}{3}$ into $g(x,y)$

$$2\left(\frac{1}{3}\right)^2 + y^2 - 1 = 0$$

$$y^2 = 1 - \frac{2}{9} = \frac{7}{9}$$

$$y = \pm \sqrt{\frac{7}{9}} = \left(\frac{1}{3} \pm \frac{\sqrt{7}}{3}\right)$$

All possible combination of (x,y) :

(x,y)	$\left(\frac{1}{\sqrt{2}}, 0\right)$	$\left(-\frac{1}{\sqrt{2}}, 0\right)$	$(0, \pm 1)$	$\left(\frac{1}{3}, \pm \frac{\sqrt{7}}{3}\right)$
$f(x,y)$ $= 4x^3 + y^2$	$4\left(\frac{1}{\sqrt{2}}\right)^3 + 0^2$ $= \sqrt{2}$	$4\left(-\frac{1}{\sqrt{2}}\right)^3 + 0^2$ $= -\sqrt{2}$	$4(0)^3 + (\pm 1)^2$ $= 1$	$4\left(\frac{1}{3}\right)^3 + \left(\pm \frac{\sqrt{7}}{3}\right)^2$ $= \frac{4}{27} + \frac{7}{9} = \frac{19}{27}$
max/min	max	min	max	max

⑧ $f(x,y) = x - 3y$;

$x^2 + 3y^2 - 16 = 0$
 $g(x,y)$

$\nabla f = \hat{i} - 3\hat{j}$

$\nabla g = 2x\hat{i} + 6y\hat{j}$

$\nabla f = \lambda \nabla g$

$\hat{i} - 3\hat{j} = \lambda(2x\hat{i} + 6y\hat{j})$

Equating Factor of Like terms:

$1 = \lambda 2x$ — (i)

$-3 = \lambda 6y$ — (j)

$\frac{1}{2x} = \lambda$

$-\frac{1}{2y} = \lambda$

Compare

$\frac{1}{2x} = -\frac{1}{2y}$

$x = -y$ — (1)

or $y = -x$ — (2)

substitute either $x = -y$ or $y = -x$ into $g(x,y)$

Substituting $y = -x$ into $g(x,y) \Rightarrow x^2 + 3(-x)^2 = 16$

Alternatively sub $x = -y$ into $g(x,y)$

$\Rightarrow (-y)^2 + 3y^2 - 16 = 0$

$4y^2 = 16$

$y = \pm 2$

$4x^2 = 16$

$x^2 = 4$

$x = \pm 2$

In eqn (1):

sub $y = 2 \rightarrow x = -2$

sub $y = -2 \rightarrow x = -(-2) = 2$

Critical pts are:

$(-2, 2), (2, -2)$

In eqn (2):

sub $x = 2 \rightarrow y = -2$

sub $x = -2 \rightarrow y = -(-2) = 2$

Critical pts are

$(2, -2), (-2, 2)$

(x, y)	$(-2, 2)$	$(2, -2)$
$f(x,y) = x - 3y$	$-2 - 3(2) = -8$	$2 - 3(-2) = 8$
max/min	min	max

16

$f(x,y)$ has maximum 8 at $(2, -2)$

$f(x,y)$ has minimum -8 at $(-2, 2)$

$$(9) f(x, y, z) = 2x + y - 2z \quad ; \quad \underbrace{x^2 + y^2 + z^2 = 4}_{g(x, y, z)}$$

$$\nabla f = 2\hat{i} + \hat{j} - \hat{k}$$

$$\nabla g = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\nabla f = \lambda \nabla g$$

$$2\hat{i} + \hat{j} - 2\hat{k} = \lambda (2x\hat{i} + 2y\hat{j} + 2z\hat{k})$$

Equating Factors of Like terms:

$$2 = \lambda 2x \quad \text{--- (i)} \quad 1 = \lambda 2y \quad \text{--- (j)} \quad -2 = \lambda 2z \quad \text{--- (k)}$$

$$\lambda = \frac{1}{x}$$

$$\lambda = \frac{1}{2y}$$

$$\lambda = \frac{-2}{2z} = -\frac{1}{z}$$

compare (i), (j)

$$\frac{1}{x} = \frac{1}{2y}$$

$$x = 2y$$

$$\boxed{y = \frac{1}{2}x}$$

compare (i), (k)

$$\frac{1}{x} = -\frac{1}{z}$$

$$x = -z$$

$$\boxed{z = -x}$$

Substitute "y" and "z" into $g(x, y)$

$$x^2 + \left(\frac{1}{2}x\right)^2 + (-x)^2 = 4$$

$$x^2 + \frac{x^2}{4} + x^2 = 4$$

$$4x^2 + x^2 + 4x^2 = 16$$

$$9x^2 = 16$$

$$x^2 = \frac{16}{9} \Rightarrow x = \pm \frac{4}{3}$$

$$x = \frac{4}{3} \Rightarrow y = \frac{1}{2}\left(\frac{4}{3}\right) = \frac{2}{3} \quad \& \quad z = -\left(\frac{4}{3}\right) = -\frac{4}{3}$$

$$\therefore (x, y, z) = \left(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3}\right)$$

$$x = -\frac{4}{3} \Rightarrow y = \frac{1}{2}\left(-\frac{4}{3}\right) = -\frac{2}{3} \quad \& \quad z = -\left(-\frac{4}{3}\right) = \frac{4}{3}$$

$$\therefore (x, y, z) = \left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$$

(x, y, z)	$(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3})$	$(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3})$
$f(x, y, z)$ $= 2x + y - 2z$	$2(\frac{4}{3}) + \frac{2}{3} - 2(-\frac{4}{3})$ $= \frac{8}{3} + \frac{2}{3} + \frac{8}{3}$ $= \frac{18}{3} = 6$	$2(-\frac{4}{3}) + (-\frac{2}{3}) - 2(\frac{4}{3})$ $= -\frac{8}{3} - \frac{2}{3} - \frac{8}{3}$ $= -\frac{18}{3} = -6$
max/min	max	min

$f(x, y, z)$ has maximum at $(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3})$

The maximum = 6

$f(x, y, z)$ has minimum at $(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3})$

The minimum = -6

⑩ $f(x, y, z) = 3x + 6y + 2z$; $g(x, y, z) = 2x^2 + 4y^2 + z^2 = 70$

$$\nabla f = 3\hat{i} + 6\hat{j} + 2\hat{k}$$

$$\nabla g = 4x\hat{i} + 8y\hat{j} + 2z\hat{k}$$

$$\nabla f = \lambda \nabla g$$

$$3\hat{i} + 6\hat{j} + 2\hat{k} = \lambda (4x\hat{i} + 8y\hat{j} + 2z\hat{k})$$

Equating factors of Like terms:

$$3 = 4x\lambda \text{ --- (i)}$$

$$6 = 8y\lambda \text{ --- (j)}$$

$$2 = 2z\lambda \text{ --- (k)}$$

$$\lambda = \frac{3}{4x}$$

$$\lambda = \frac{6}{8y} = \frac{3}{4y}$$

$$\lambda = \frac{1}{z}$$

compare (i), (j)

$$\frac{3}{4x} = \frac{3}{4y}$$

$$\frac{1}{x} = \frac{1}{y}$$

$$\boxed{y = x}$$

compare (i), (k)

$$\frac{3}{4x} = \frac{1}{z}$$

$$3z = 4x$$

$$\boxed{z = \frac{4x}{3}}$$

Substitute "y" and "z" into $g(x, y)$

$$2x^2 + 4x^2 + \left(\frac{4x}{3}\right)^2 = 70$$

$$2x^2 + 4x^2 + \frac{16x^2}{9} = 70$$

$$18x^2 + 36x^2 + 16x^2 = 630$$

$$70x^2 = 630$$

$$x^2 = 9$$

$$x = \pm 3$$

$$\text{Sub } x=3 \Rightarrow y=x=3 \quad \& \quad z = \frac{4x}{3} = \frac{4(3)}{3} = 4$$

$$\therefore (x, y, z) = (3, 3, 4)$$

$$\text{Sub } x=-3 \Rightarrow y=x=-3 \quad \& \quad z = \frac{4x}{3} = \frac{4(-3)}{3} = -4$$

$$\therefore (x, y, z) = (-3, -3, -4)$$

(x, y, z)	$(3, 3, 4)$	$(-3, -3, -4)$
$f(x, y, z)$ $= 3x + 6y + 2z$	$3(3) + 6(3) + 2(4)$ $= 35$	$3(-3) + 6(-3) + 2(-4)$ $= -35$
max/min	max	min

$f(x, y, z)$ has maximum at $(3, 3, 4)$

The maximum = 35

$f(x, y, z)$ has minimum at $(-3, -3, -4)$

The minimum = -35

⑪ $f(x,y,z) = xyz$, $g(x,y,z) = x^2 + y^2 + z^2 = 1$

$\nabla f = yz\hat{i} + xz\hat{j} + xy\hat{k}$ $\nabla g = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$

$\nabla f = \lambda \nabla g$

$yz\hat{i} + xz\hat{j} + xy\hat{k} = \lambda (2x\hat{i} + 2y\hat{j} + 2z\hat{k})$

Equating factors of like terms

$yz = \lambda 2x$ — (i)

$xz = 2y\lambda$ — (j)

$xy = 2z\lambda$ — (k)

$\lambda = \frac{yz}{2x}$

$\lambda = \frac{xz}{2y}$

$\lambda = \frac{xy}{2z}$

compare (i), (j), (k)

$\frac{yz}{2x} = \frac{xz}{2y} = \frac{xy}{2z} \Rightarrow \frac{yz}{x} = \frac{xz}{y} = \frac{xy}{z}$

(i), (j)

$\frac{yz}{x} = \frac{xz}{y}$

$\frac{y}{x} = \frac{x}{y}$

$y^2 = x^2$

sub $x = \pm \frac{1}{\sqrt{3}}$ $\therefore y^2 = (\pm \frac{1}{\sqrt{3}})^2 = \frac{1}{3}$

$y = \pm \frac{1}{\sqrt{3}}$

$\frac{yz}{x} = \frac{xy}{z}$

$\frac{z}{x} = \frac{x}{z}$

$x^2 = z^2$

sub $x = \pm \frac{1}{\sqrt{3}}$ $\therefore z^2 = (\pm \frac{1}{\sqrt{3}})^2 = \frac{1}{3}$

$z = \pm \frac{1}{\sqrt{3}}$

Substitute y^2, z^2 into $g(x,y)$:

$x^2 + (x^2) + (x^2) = 1$

$3x^2 = 1$

$x = \pm \frac{1}{\sqrt{3}}$

2 variables:
 $2^2 = 4$ outcomes
3 variables:
 $2^3 = 8$ outcomes
 n variables:
 2^n outcomes.

All possible combination of $(x,y,z) = 2^3 = 8$ many

(x,y,z)	$(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$	$(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$
$f(x,y,z) = xyz$	$\frac{1}{3\sqrt{3}}$	$-\frac{1}{3\sqrt{3}}$	$-\frac{1}{3\sqrt{3}}$	$-\frac{1}{3\sqrt{3}}$
max/min	maximum	minimum	minimum	minimum
(x,y,z)	$(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$	$(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$	$(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$
$f(x,y,z) = xyz$	$-\frac{1}{3\sqrt{3}}$	$\frac{1}{3\sqrt{3}}$	$\frac{1}{3\sqrt{3}}$	$\frac{1}{3\sqrt{3}}$
max/min	minimum	maximum	maximum	maximum

The maximum is $\frac{1}{3\sqrt{3}}$ and it occurs at

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right).$$

The minimum is $-\frac{1}{3\sqrt{3}}$ and it occurs at

$$\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right).$$

(12) $f(x, y, z) = x^4 + y^4 + z^4$; $g(x, y, z) = x^2 + y^2 + z^2 = 1$

$$\nabla f = 4x^3\hat{i} + 4y^3\hat{j} + 4z^3\hat{k} \quad \nabla g = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\nabla f = \lambda \nabla g$$

$$4x^3\hat{i} + 4y^3\hat{j} + 4z^3\hat{k} = \lambda (2x\hat{i} + 2y\hat{j} + 2z\hat{k})$$

$$4x^3 = \lambda 2x \quad \text{--- (i)}$$

$$4y^3 = \lambda 2y \quad \text{--- (j)}$$

$$4z^3 = \lambda 2z \quad \text{--- (k)}$$

$$\rightarrow 4x^3 - \lambda 2x = 0$$

$$2x(2x^2 - \lambda) = 0$$

$$2x = 0$$

$$x = 0$$

$$2x^2 = \lambda \quad \text{--- (i')}$$

$$2y = 0$$

$$2y(2y^2 - \lambda) = 0$$

$$2y^2 = \lambda \quad \text{--- (j')}$$

$$y = 0$$

$$\rightarrow z = 0 \text{ \& } 2z^2 = \lambda \quad \text{--- (k')}$$

$$(x, y, z) = (0, 0, 0)$$

comparing (i'), (j'), (k')

$$\Rightarrow 2x^2 = 2y^2 = 2z^2$$

$$\Rightarrow x^2 = y^2 = z^2$$

Substitute $y^2 = x^2$, $z^2 = x^2$ into $g(x, y, z)$:

$$x^2 + (x^2) + (x^2) = 1$$

$$3x^2 = 1$$

$$x = \pm \frac{1}{\sqrt{3}}$$

$$\because y^2 = x^2 \quad \because y^2 = \left(\pm \frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3} \Rightarrow y = \pm \frac{1}{\sqrt{3}}$$

$$\because z^2 = x^2 \quad \because z^2 = \left(\pm \frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3} \Rightarrow z = \pm \frac{1}{\sqrt{3}}$$

Total number of outcomes = $2^3 = 8$ $\because (x, y, z) \leftrightarrow 3$ variables

(x, y, z)	$f(x, y, z) = x^4 + y^4 + z^4$	max/min
$(0, 0, 0)$	0	—
$(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$(\frac{1}{\sqrt{3}})^4 + (\frac{1}{\sqrt{3}})^4 + (\frac{1}{\sqrt{3}})^4$ $= \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{3}{9} = \frac{1}{3}$	maximum
$(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$\frac{1}{3}$	"
$(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$\frac{1}{3}$	"
$(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$	$\frac{1}{3}$	"
$(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$	$\frac{1}{3}$	"
$(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$\frac{1}{3}$	"
$(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$	$\frac{1}{3}$	"
$(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$	$\frac{1}{3}$	"